6 Interface characteristics

6.1 Bond of embedded steel reinforcement

Bond is the term used to denote the interaction and transfer of force between reinforcement and concrete. Bond influences performance of concrete structures in several ways. At the serviceability limit state, bond influences width and spacing of transverse cracks, tension stiffening and curvature. At the ultimate limit state, bond is responsible for strength of end anchorages and lapped joints of reinforcement, and influences rotation capacity of plastic hinge regions.

6.1.1 Local bond–slip relationship

6.1.1.1 Local bond stress–slip model, ribbed bars

Under well-defined conditions, it is possible to consider that there is an average "local bond" versus "local slip" relationship, for short anchorage lengths, statistically acceptable.

This section covers anchorage of ribbed reinforcing bars which satisfy the requirements for classification as "high bond" in accordance with requirements of section 5.2.

For monotonic loading the reference value τ_b of the bond stresses between concrete and reinforcing bar for pull-out and splitting failure can be calculated as a function of the relative displacement s parallel to the bar axis as follows (Figure 6.1-1):

$$\tau_b = \tau_{bmax} (s/s_1)^{\alpha} \quad \text{for} \quad 0 \le s \le s_1 \tag{6.1-1}$$

$$\tau_b = \tau_{bmax} \quad \text{for} \quad s_1 \le s \le s_2 \tag{6.1-2}$$

$$\tau_b = \tau_{bmax} - (\tau_{bmax} - \tau_{bf}) (s\text{-}s_2)/(s_3\text{-}s_2) \quad \text{for} \quad s_2 \le s \le s_3 \tag{6.1-3}$$

$$\tau_b = \tau_{bf} \quad \text{for} \quad s_3 < s \tag{6.1-4}$$

where the parameters are given in Table 6.1-1

Explicit consideration of the influence of bond on performance at the serviceability limit state and on rotation capacity is not normally required for members reinforced with conventional steel bars. The influence is instead represented by a modification of the tension stiffening from the surrounding concrete.

The bond stress–slip relationship depends on a considerable number of influencing factors including rib geometry (relative rib area), concrete strength, position and orientation of the bar during casting, state of stress, boundary conditions and concrete cover. The parameters given in Table 6.1-1 are valid for ribbed reinforcing steel with a relative rib area $f_r \ge f_{r,min}$ according to relevant international standards – see section 5.2.

The bond stress–slip curves for confined and unconfined concrete presented in Figure 6.1-1 can be considered applicable as an average formulation for a broad range of cases. Further reliability handling would be needed to derive design bond stress–slip curves.

The ascending part refers to the stage in which the ribs penetrate into the mortar matrix, characterized by local crushing and micro-cracking. A sustained plateau occurs only for confined concrete, during which advanced crushing and shearing off of the concrete between the ribs takes place. This level represents a residual bond capacity which is maintained only where a large concrete cover, dense transverse reinforcement or transverse compression is present to keep a certain degree of integrity intact. The descending branch refers to the reduction of bond resistance as concrete corbels between the ribs are sheared off. In the case of unconfined concrete splitting, failure occurs which is reflected by a sudden drop in bond stress before a constant residual level is reached. The peak value of bond strength in a splitting failure mode is denoted $\tau_{bu,split}$ – see Figure 6.1-1 and Eq. (6.1-5).

With regard to the development of bond stresses, the following considerations apply:

Reinforcement and concrete have the same strain ($\varepsilon_s = \varepsilon_c$) in those areas of the structure where the steel is in compression (outside eventual load introduction areas) and in those areas where the steel is in tension in uncracked parts of the structure.

In cracked cross-sections tension forces are transferred across the crack by the reinforcing steel. In general, the absolute displacements of the steel u_s and of the concrete u_c adjacent to a crack are different. Differences in displacements are similarly found along the transmission length l_{bpt} of pretensioned prestressed concrete members.

Due to the relative displacement $s = u_s - u_c$ bond stresses are generated between concrete and reinforcing steel or prestressing tendons. For $s < s_1$, the magnitude of these bond stresses depends predominantly on the slip s, but is influenced by the surface of the steel, concrete strength f_{cm}, the position of the reinforcing steel during concreting and by confinement from concrete cover, secondary reinforcement and the state of stress in the concrete around the bar. Between cracks or along the transmission length

l_{bpt} a part of the tension force of the reinforcing steel at the cracked section is transferred into the concrete by bond (tension stiffening effect). The local change in relative displacement is characterized by the strain difference $ds/dx = \varepsilon_s - \varepsilon_c$.

Depending on the selection of the coefficient α ($0 \le \alpha \le 1$) in Eq. (6.1-1) all usual forms of a bond stress–slip relationship can be modelled, starting from a bond characteristic with a constant stress ($\alpha = 0$) up to a bond stress–slip relationship with linear increasing bond stress ($\alpha = 1$).

The parameters from Table 6.1-1 to be applied are dependent primarily on the failure mode, pull-out or splitting. Further distinctions are made to account for differences in bond conditions within each failure mode.

The rationale underpinning the following rules is given in ***fib*** Bulletin "Bond and Anchorage of Reinforcement: Background to ***fib*** Model Code 2010".

Table 6.1-1: Parameters defining the mean bond stress–slip relationship of ribbed bars (according to Eqs. (6.1-1)–(6.1-4))

	1	2	3	4	5	6
	Pull-out (PO)		**Splitting (SP)**			
	$\varepsilon_s < \varepsilon_{s,y}$		$\varepsilon_s < \varepsilon_{s,y}$			
	Good bond cond.	**All other bond cond.**	**Good bond cond.**		**All other bond cond.**	
			Unconfined	**Stirrups**	**Unconfined**	**Stirrups**
τ_{bmax}	$2.5\sqrt{f_{cm}}$	$1.25\sqrt{f_{cm}}$	$2.5\sqrt{f_{cm}}$	$2.5\sqrt{f_{cm}}$	$1.25\sqrt{f_{cm}}$	$1.25\sqrt{f_{cm}}$
$\tau_{bu,split}$	—	—	$7.0\cdot\left(\frac{f_{cm}}{25}\right)^{0.25}$	$8.0\cdot\left(\frac{f_{cm}}{25}\right)^{0.25}$	$5.0\cdot\left(\frac{f_{cm}}{25}\right)^{0.25}$	$5.5\cdot\left(\frac{f_{cm}}{25}\right)^{0.25}$
s_1	1.0 mm	1.8 mm	$s(\tau_{bu,split})$	$s(\tau_{bu,split})$	$s(\tau_{bu,split})$	$s(\tau_{bu,split})$
s_2	2.0 mm	3.6 mm	s_1	s_1	s_1	s_1
s_3	c_{clear} [1)]	c_{clear} [1)]	$1.2s_1$	$0.5c_{clear}$ [1)]	$1.2s_1$	$0.5c_{clear}$ [1)]
a	0.4	0.4	0.4	0.4	0.4	0.4
τ_{bf}	$0.40\tau_{max}$	$0.40\tau_{max}$	0	$0.4\tau_{bu,split}$	0	$0.4\tau_{bu,split}$

[1)] c_{clear} is the clear distance between ribs

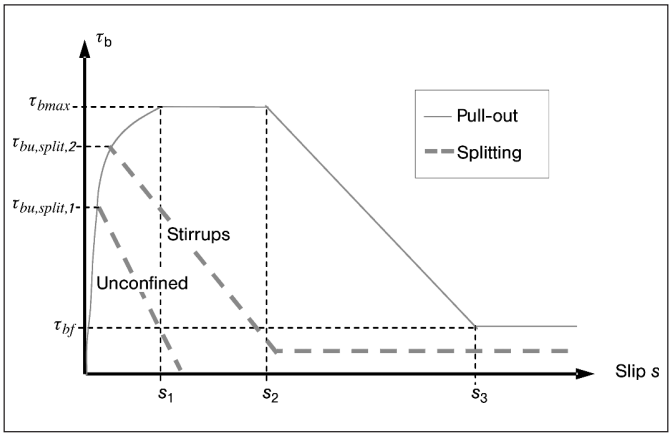


Figure 6.1-1: Analytical bond stress–slip relationship (monotonic loading). $\tau_{bu,split,1}$ and $\tau_{bu,split,2}$ denote peak local bond resistance in the absence ($K_{tr} = 0$) and presence, respectively, of confining stirrups

The values in Table 6.1-1, columns 1 and 2 (pull-out failure) are valid for well-confined concrete (concrete cover ≥ 5Ø, clear spacing between bars ≥ 10Ø), or suitable confining reinforcement.

The values in Table 6.1-1, columns 3 to 6 (splitting failure) are derived from Eq. (6.1-5), which is derived from Eq. (6.1-19) by setting $l_b/Ø = 5$, assuming a uniform bond stress over this length and evaluating $\tau_{bu,split}$ for Ø = 25 mm, $c_{max}/c_{min} = 2.0$, $c_{min} = Ø$ and $K_{tr} = 0.02$ in the case where bars are confined by stirrups or $K_{tr} = 0$ where no stirrups are provided.

$$\tau_{bu,split} = \eta_2 6.5 \left(\frac{f_{cm}}{25}\right)^{0.25} \left(\frac{25}{Ø}\right)^{0.2} \left[\left(\frac{c_{min}}{Ø}\right)^{0.33} \left(\frac{c_{max}}{c_{min}}\right)^{0.1} + k_m K_{tr}\right] \tag{6.1-5}$$

where:

η_2 = 1.0 for good bond conditions;
= 0.7 for all other bond conditions (see subsection 6.1.3.2 for definition of bond conditions);

f_{cm} is the mean cylinder concrete compressive strength [N/mm^2];

Ø is the diameter of the anchored bar considered [mm];

c_{min} = min{$c_s/2$, c_x,c_y} (Figure 6.1-2);

c_{max} = max{$c_s/2$, c_x} (Figure 6.1-2).

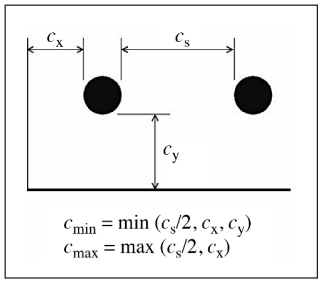


Figure 6.1-2: Notation for bar spacing and cover: straight bars

k_m represents the efficiency of confinement from transverse reinforcement, and has a value of 12 where bars are confined inside a bend of links passing round the bar of at least 90°. Where no confining reinforcement is provided between bars and the nearest face, $k_m = 0$, as shown in Figure 6.1-3.

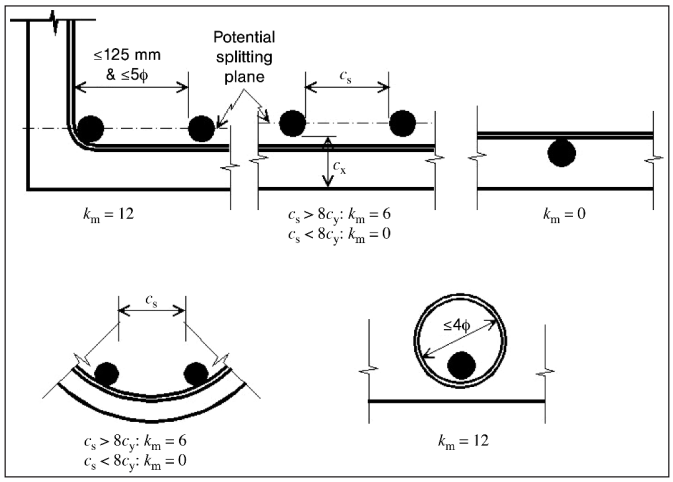


Figure 6.1-3: Confinement coefficients for transverse reinforcement

$$K_{tr} = n_t A_{st}/(n_b \, Ø \, s_t) \leq 0.05 \qquad (6.1\text{-}6)$$

where:

n_t is the number of legs of confining reinforcement crossing a potential splitting failure surface at a section;

A_{st} is the cross-sectional area of one leg of a confining bar [mm^2];

s_t is the longitudinal spacing of confining reinforcement [mm];

n_b is the number of anchored bars or pairs of lapped bars in the potential splitting surface.

The bond stress–slip curve is considerably influenced by reinforcement yielding, by transverse pressure and cracking along the bar, and by cyclic, repeated or sustained loading.

6.1.1.2 Influence of transverse cracking

For those parts of the reinforcing bar which are at a distance $x \leq 2 \, Ø$ from a transverse crack, the bond stress τ is to be reduced by the factor λ where:

$$\lambda = 0.5 \, x/Ø \leq 1 \qquad (6.1\text{-}7)$$

6.1.1.3 Influence of yielding, transverse stress and longitudinal cracking and cyclic loading

The bond stress according to Eqs. (6.1-1)–(6.1-4) should be modified by the factors $_{y,}$ $_{p,tr}$ and $_{cr}$ and $_{cyc}$ in case of bar yielding, transverse pressure and cracking parallel to the bar axis and cyclic loading respectively:

$$\tau_{b,m} = \tau_{b0} \cdot \Omega_y \cdot \Omega_{p,tr} \cdot \Omega_{cr} \cdot \Omega_{cyc} \qquad (6.1\text{-}8)$$